



The Trajectory, Chronicity, and Etiology of Problem Gambling: A Synthesis of Longitudinal Research

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Abstract

Purpose of Review To summarize longitudinal research findings on the trajectory, chronicity, and etiology of problem gambling.

Recent Findings Problem gambling is not usually progressive. Rather, most people with early problems reduce their subsequent involvement and sustain this reduction. Problem gambling is also not usually chronic and unremitting. A minority do have a sustained chronic course, particularly more severe cases. It is also enduring in that relapse and risk of relapse remains high in anyone who has ever had the condition. However, the condition most typically remits, with some people having sustained remission and others having repeated relapses and remissions. In terms of etiology, up to 20 variables additively contribute to the initial onset of problem gambling with different combinations of variables being present in different people. That said, intensive gambling involvement, gambling fallacies, addiction-propensity, a social group that regularly gambles, and impulsivity tend to be particularly important factors.

Summary Problem gambling has a biopsychosocial etiology with a trajectory and chronicity very similar to substance use disorder.

Keywords Problem gambling · Longitudinal · Etiology · Trajectory · Chronicity

Introduction

The correlates of problem gambling are well known [1–4]. However, the etiological role of many of these correlates is uncertain, as they may have appeared as a consequence of problem gambling or developed coincidentally with it, rather than having causal role. *Longitudinal research* is required to help disentangle their sequential chronology and potential causal relationship to gambling-related harm. Longitudinal research is also essential in understanding the trajectory and chronicity of problem gambling over time.

Longitudinal studies of gambling were largely non-existent prior to the early 2000s. However, it was the absence

of this type of research that spurred their appearance. Currently, over 50 longitudinal studies of gambling now exist. Most of these studies are identified and described in the prior reviews of Dowling et al. [5] and Williams et al. [6]. Additional studies were identified for the present review by a systematic search of Medline, PsycInfo, and Google Scholar using the terms gambling, problem gambling, disordered gambling, pathological gambling, and compulsive gambling in combination with the terms longitudinal, cohort, and prospective.

Most longitudinal studies have somewhat limited generalizability due to either: (a) Only studying the onset of gambling, but not problem gambling; (b) A focus on a very specific demographic (e.g., youth, elderly, casino employees); (c) Having a very small sample size and/or a very small number of people who developed gambling problems; (d) Conducting the study over a very short time span and/or a small number of assessment periods; (e) Only investigating the *concurrent* rather than the *prospective* relationship of variables over time with problem gambling; or (f) Assessing only a small subset of etiologically relevant variables.

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However, in recent years there have been seven large scale longitudinal studies of problem gambling conducted in five different countries that utilized large jurisdictionally representative samples, multiple assessment periods over multiple years, and a more comprehensive assessment of etiologically relevant variables¹:

- Sweden: Swedish Longitudinal Gambling Study (SWELOGS) [7–9] (2008–2015; 4 waves; $n = 15,000$ at Baseline).
- Australia: Victorian Gambling Study (VGS) [10, 11] (2008–2011; 4 waves; $n = 15,000$ at Baseline).
- New Zealand: New Zealand National Gambling Study (NZNGS) [12, 13] (2012–2018; 4 waves; $n = 6,251$ at Baseline).
- United States: Massachusetts Gambling Impact Cohort (MAGIC) [14] (2013–2019; 5 waves; $n = 3,139$ at Baseline).
- Canada: Leisure, Lifestyle, and Lifecycle Project (LLLP) in Alberta [15] (2006–2011; 4 waves; $n = 1,818$ at Baseline); Quinte Longitudinal Study (QLS) in Ontario [6] (2006–2011; 5 waves; $n = 4,121$ at Baseline); and the Alberta Gambling Research Institute National Project (ANP-1) [16] (2018–2019; 1 wave; $n = 10,199$ at Baseline).

Collectively, these large-scale comprehensive studies as well as the more circumscribed longitudinal studies have provided a wealth of information on the trajectory, stability and etiology of problem gambling. The purpose of the present paper is to summarize these results in a narrative review with a focus on the most recent findings.

Trajectory of Problem Gambling

Historically, it was a common belief that problem gambling was a progressive condition whereby symptoms inevitably worsen once they emerge [17, 18]. Further evidence is seen in its designation as ‘pathological gambling’ by the American Psychiatric Association in the 1980 Diagnostic and Statistical Manual of Mental Disorders (DSM)-III and 1993 DSM-IV (pathological meaning ‘disease-like’, with diseases typically having a progressive course). Gamblers Anonymous has always contended that it is a progressive condition with their website stating that “... gamblers of our type are in the grip of a progressive illness. Over any considerable period of time we get worse, never better.” [19].

However, this belief was largely based on the experience of clinicians and people presenting to treatment as well as the 20th century embrace of the ‘disease model’ of addiction [20, 21]. The first *empirical* review of this assumption was undertaken by LaPlante et al. in 2008 [22] who reviewed data from five longitudinal studies: [23–27]. Depending on the study and the severity of symptomatology it was found that between 54.4 and 90.9% of people with subclinical problem gambling symptomatology (i.e., people who reported symptoms below the level needed to be designated as a problem gambler) were found to have *improved over time* compared to only 0.0–13.6% who worsened.

All the recent large-scale longitudinal gambling studies have subsequently confirmed this finding. These studies have found that not only is subclinical problem gambling symptomatology not progressive, but it is actually a *poor predictor of future problem gambling*. As an illustrative example, in SWELOGS among the 307 people in the low-risk category (score of 1–2) on the Problem Gambling Severity Index (PGSI) [28], 66.1% had no symptoms 12 months later and for the 116 people in the moderate risk category (PGSI 3–7), 56.0% had no symptoms 12 months later [29]. Similarly, in the QLS only 14.7% of the 482 individuals in the at-risk category of the Problem and Pathological Gambling Measure (PPGM) [30] became a problem gambler in the subsequent four years, with the vast majority instead reverting back to recreational gambling [6]. Most recently, in the Canadian national study (ANP-1), of the 1,388 people in the PPGM at-risk category at baseline, only 20.4% progressed to problem gambling at the one-year follow-up [16]. Thus, in summary, *instead of problem gambling symptoms inevitably leading to progressively more problems in the future, it appears that early problems typically dissipate, potentially serving as a ‘wake-up call’ causing most people to reduce their subsequent involvement*. It is interesting to note that longitudinal studies of alcohol and drug use have found a very similar pattern [25, 31, 32].

Chronicity of Problem Gambling

Historically, problem gambling was also thought to be unremitting and enduring, as evidenced by its designation as ‘pathological gambling’ and no time frame being specified for the diagnosis in the DSM-III or DSM-IV, implying that this was a lifetime disorder. Also, as mentioned earlier, Gamblers Anonymous still uses the term ‘compulsive gambling’, with their website stating, “compulsive gambling is an illness, progressive in its nature, which can never be cured...” [19].

However, longitudinal investigations in the early 2000s found evidence of considerable instability over time and some cases of long-term remission. The LaPlante et al. 2008 review of five longitudinal studies found that among people

¹ There are two additional large scale longitudinal studies without a gambling focus, but which have included an assessment of gambling or problem gambling: Pacific Islands Families Study, New Zealand Dunedin Multidisciplinary Health and Development Study.

with clinical levels of problem gambling that *between 49.9 and 88.9% subsequently reduced their symptomatology to subclinical levels or lower* [22].

The more recent longitudinal studies have now largely confirmed these findings:

- Luce et al. [33] conducted annual telephone assessments over three years of an initial sample of 179 people from Quebec, including 17 people with PGSI 8+ (severe problem gambling) at baseline. At the third assessment, 64.7% were still in the same category, but only 41.2% were PGSI 8+ at all three waves.
- Black et al. [34] followed 105 individuals with DSM-IV scores of five or higher in Iowa every six months for roughly 2.6 years and found that only 45.3% of younger and 49.3% of older individuals were continuously scoring in the 5+ range.
- In SWELOGS, Slecza & Romild [29] found that for the 16 people with PGSI 8+, 87.5% still had some problems 12 months later but only 20.0% had problems five years later.
- In VGS, 55% of the 42 people scoring PGSI 8+ were PGSI 8+ in the fourth assessment and 48% were PGSI 8+ in all four waves [10].
- In MAGIC, one year was the modal duration of PPGM Problem Gambling, occurring in 50.3%. Of the 53 Problem Gamblers in Wave 1, 47.2% were in this category in Wave 5, but only 18.9% were in this category in all five waves. Remittance rates tended to be high, with most having at least one year of remission over the five waves. However, relapse rates were also quite high. Of the 27 people that had remitted in Wave 2, 33.3% had relapsed by Wave 3, 4, or 5 [14].
- In LLLP, one Wave (one year) was also the modal duration of PGSI 5+, occurring in 47.4% of the 57 problem gamblers identified in Waves 1 to 3. Of the 23 problem gamblers in Wave 1, 60.9% were in this category in Wave 4, but only 43.5% being in this category in all four waves. Of the 9 people that had remitted in Wave 2, 33.3% had relapsed by Wave 3 or 4 [15].
- In QLS, one Wave (one year) was also the modal duration of PPGM problem gambling, occurring in 50.8% of the 236 problem gamblers identified in Waves 1 to 4. Of the 109 problem gamblers in Wave 1, 33.0% were in this category in Wave 5, but only 17.4% being in this category in all five waves. Of 58 people that had remitted in Wave 2, 41.8% had relapsed by Wave 3, 4, or 5 [6].

Potential limitations of some of the above studies concern the relatively small number of problem gamblers that were followed and the very similar time frames involved (i.e., typically an annual assessment conducted over three to five years). However, the sample size issue was addressed

in the Canadian ANP-1, where 65.0% of the 1,346 PPGM problem gamblers in Wave 1 were still problem gamblers in Wave 2, one year later [16]. The time frame issue is addressed by two different studies. Hofmeyr et al. [35] and Dellis et al. [36] conducted six face-to-face assessments over a 15-month period in 2010–2011 of a South African sample that included 93 PGSI 8+ problem gamblers and found that only 44.2% were in the same category at the next assessment. Slutske et al. [37] examined data from the New Zealand Dunedin study that assessed people at ages 21, 32, and 45. Of the 125 problem gamblers at age 21 only 56 still met criteria for problem gambling at age 32 and only 29 at age 45.

Thus, in summary, it is clear that problem gambling is not inevitably chronic and unremitting. Rather, *(a) only a portion of people with gambling problems have unremitting problem gambling; and (b) there is considerable variability between individuals, with some people having a more chronic course, other people remitting after a single episode, and other people transitioning in and out of the condition. That said, problem gambling is an enduring condition in the sense that relapse and risk of relapse remains high in anyone who has ever had the condition.* Coincident with these above findings was a change in how this condition was described and assessed. Beginning in 2007 ‘problem gambling’ replaced ‘compulsive gambling’ and ‘pathological gambling’ as the most common used term in the literature (based on the number of ‘hits’ on Google Scholar).² In addition, ‘pathological gambling’ was changed to ‘gambling disorder’ in the DSM-5 [38]. The instruments utilized to assess problem gambling also began to routinely employ a 12-month assessment frame. This began with the South Oaks Gambling Screen (‘SOGS-Revised’) in the 1990s [39], the National Opinion Research Center DSM-IV Screen for Gambling Problems (NODS) in 1999 [40], the Victorian Gambling Screen in 1999 [41], the Problem Gambling Severity Index in 2001 [28], the Problem and Pathological Gambling Measure in 2010 [30, 42], and the DSM-5 in 2013 [38], which also introduced specifiers of ‘episodic or persistent’ and ‘early or sustained remission’.

The above chronicity findings parallel what has been found for substance use disorder [43–46]. These parallels have been further confirmed in a couple of head-to-head studies. Shaffer & Hall [25] in their three-year prospective study of 6,067 U.S. casino employees found very similar levels of instability between problem gambling and problem drinking. Gooding et al. [47] also found very similar

² ‘Problem gambling’ has continued to be the most common term on Google Scholar up to the present time relative to ‘disordered gambling’, ‘compulsive gambling’, ‘pathological gambling’, and ‘addicted gambler/gambling’.

levels of chronicity between gambling disorder and substance use disorder in the QLS, with the modal duration of both conditions being one wave and the average number of assessment periods in the clinical state being 1.7 for substance use disorder and 2.2 for gambling disorder (over five waves).

While it is clear that problem gambling has a variable course, it is instructive to understand why it is chronic in some cases and transitory in others, as well as why chronicity was higher in some of the above studies and lower in others. *One of the primary determinants has to do with severity.* There are three aspects of severity: the level and extensiveness of the problem gambling symptomatology; the duration of the existing episode; and whether significant comorbidities are present. All three appear to be important. As illustration of the first aspect, higher chronicity was found in longitudinal studies that have used PGSI 8+ as the demarcation of problem gambling compared to studies that utilized PGSI 5+ or the PPGM (roughly equivalent to a PGSI 5+). Studies that have examined how individual differences in severity relate to chronicity have also found higher individual PGSI, PPGM, or DSM scores predict episodes that persist over two or more consecutive assessment periods, and/or higher rates of relapse [5–7, 10, 14–16, 33, 36, 48–50]. The second and third aspects of severity have been less well studied. However, two studies have established that every consecutive year of having the condition predicts decreased likelihood of remission in the subsequent year [15, 16]. Similarly, a lifetime history of problem gambling was the strongest predictor of relapse in the MAGIC study [14]. Finally, Gooding et al. [47] showed that the chronicity of both gambling disorder and substance use disorder was higher when the person had multiple addictions (e.g., gambling disorder plus substance use disorder).

Etiology of Problem Gambling

There have been roughly 20+ longitudinal studies specifically investigating variables predictive of problem gambling. As mentioned earlier, many of these studies have limited generalizability either because of using a demographically circumscribed sample; having a small number of people with problem gambling in the study; using only concurrent rather than prospective analyses; only assessing a subset of variables of etiological importance; and/or not using multivariate analytic approaches that would establish whether bivariate associations with problem gambling are an artifact of the variable's strong association with other variables having a greater direct causal relationship.

Taking these issues into account, as well as previous earlier reviews of the etiological predictors [5, 6] and recent studies that have addressed the above issues (e.g [14, 16]), some conclusions are warranted:

The first conclusion is that *up to 20 variables from different domains collectively and additively contribute to the development of problem gambling.* No study has found there to be any individual variable or small subset of variables that strongly predicts problem gambling. For example, MAGIC utilized 105 independent variables (chosen because of their established bivariate association with problem gambling) and found 19 to have a concurrent multivariate relationship, 17 with a future multivariate relationship, and 13 to have both [14].³ The Canadian national study (ANP-1) utilized a comparable comprehensive set of 108 independent variables and found there to be 23 multivariate predictors of concurrent problem gambling, 14 multivariate predictors of future problem gambling, and 11 multivariate predictors of both [16].

Furthermore, the evidence shows there to be *different pathways and different combinations of game play, personality, cognitive, family history, mental health, demographic, contextual, etc. variables that lead to problem gambling,* aligning with a general biopsychosocial understanding of addiction [51, 52]. Although the popular Pathways Model of Problem Gambling [53, 54] proposes only three pathways⁴, empirical support for this model consists primarily of cross-sectional identification of problem gambler subtype groupings utilizing a small subset of variables pertinent to the Pathways Model [55, 56]. Subgroup identification using a much broader set of etiologically relevant variables (based on datasets available to the first author of the present manuscript, RW) shows additional subgroups (e.g., skill-based gamblers; gamblers with a high level of gambling fallacies). Furthermore, these datasets show that while people with prominent mental health issues or impulsivity/antisocial traits are important subgroups, they constitute the minority of problem gamblers and virtually all members of these subgroups have many additional risk factors that additively contributed to the subsequent appearance of problem gambling. (As the large majority of people with only mental health problems and/or impulsivity do not progress to problematic gambling). One of these additional risk factors is intensive gambling involvement, present in the vast majority (i.e., virtually all problem gamblers are 'behaviourally conditioned' to some extent).

Another important conclusion is that the multivariate risk factors for problem gambling have a *hierarchy of importance*

³ As expected, almost all of the 105 independent variables also showed a bivariate relationship with problem gambling.

⁴ A 'behaviorally conditioned' etiological pathway for individuals with few mental health or personality risk factors; an 'emotionally vulnerable' pathway for individuals with high levels of anxiety, depression, and child abuse; and an 'antisocial impulsivist' pathway for people with high levels of risk taking, impulsivity and antisocial traits.

due to their pervasiveness amongst future problem gamblers; and/or their presence among future problem gamblers relative to future nonproblem gamblers; and/or their cross-cultural predictive power. In general, *the strongest multivariate predictors are ‘gambling-related’ variables*:⁵

- *Current problem gambling status* is the strongest predictor. Similarly, reporting *any past history of problem gambling* is also very strong. Although most studies have not included these variables as predictors, all studies that have included them have confirmed them to be the strongest or among the strongest predictors [6, 10, 14–16, 23, 57, 58]. In addition, as documented earlier, *problem gambling severity* is also a strong predictor of chronicity.
- *Intensive gambling involvement* (as measured by magnitude of gambling losses, gambling frequency, time spent gambling, and number of formats involved in) is also a very strong predictor [5, 6, 14–16, 59].⁶ While commonsensical, it confirms that intensive gambling involvement is the final common pathway and immediate antecedent to gambling-related problems. Intensive gambling involvement, in turn, is related to *rating gambling as an important leisure activity*, which is also a very strong predictor [6, 14, 16].
- *Regular involvement in continuous forms of gambling and/or online gambling* are strong predictors. Continuous types of gambling such as electronic gambling machines (EGMs) and casino table games theoretically have a strong association with harm because of their higher frequency/amount of expenditure⁷ combined with a higher frequency of reinforcement [1, 4, 60–64].⁸ This higher risk has been confirmed in several longitudinal studies [6, 7, 14–16, 66] and is a particularly prominent etiological factor in people without mental health problems and/or impulsivity. To a lesser extent, online gambling has been shown to be associated with problem gambling due to its convenience, 24-hour availability, ability to play while intoxicated, and provision of continuous forms [61, 67, 68]. Some longitudinal studies have confirmed this

association [6, 14, 68]. The weaker association in a few studies is partly because the most common type of online gambling in some jurisdictions is the online purchase of lottery tickets.

- *Gambling fallacies* have a well established theoretical relationship with problem gambling [69] and have been shown to be predict future gambling problems in almost all of the recent longitudinal studies [6, 14–16, 48, 70, 71].
- Social environment is predictive of future gambling problems as measured by ‘*portion of social group with who gamble regularly*’, ‘*portion of social group who have gambling problems*’ as well as ‘*family history of problem gambling*.’ Social group has previously been documented as an important cross-sectional contributor [72, 73] likely through normalizing and facilitating excessive involvement [74]. Social group involvement has also been a future predictor in recent longitudinal studies that have included this variable [6, 14, 16]. Family history of gambling problems may similarly speak to modelling/normalization and/or genetics.

Beyond gambling-related variables, there are several other reasonably strong multivariate predictors of concurrent and/or future problem gambling status. Listed in order of importance, they are:

- *Impulsivity* is a robust predictor in many cross-sectional and longitudinal studies [2, 5–7, 10, 12, 15, 16, 75–80]. Impulsivity also bears a strong association with *antisocial traits*, which have also been implicated in longitudinal research [5, 14, 57–59, 81, 82].
- *Addiction propensity*. Having a current or past history of either substance use disorder (including regular tobacco use) or a non-gambling behavioural addiction is a reliable predictor [1, 4–7, 10, 14, 15]. This association likely reflects some shared genetic vulnerability to addiction [83–85].
- *Mental health disorders*. Depression is a reliable predictor of future gambling problems [5, 6, 14, 15, 25, 48, 57, 58, 79, 82, 86, 87]. However, a range of other mental health disorders (especially anxiety disorders such as post-traumatic stress) have also been significant predictors depending on the specific disorder and the specific study [6, 10, 14, 16]. Related to mood disorders, ‘*gambling to escape*’ is a reliable multivariate motivational predictor of concurrent and future problem gambling [6, 16, 88].
- *Lower household income* has been identified as a predictor in several cross-sectional studies [3, 4, 89, 90]. It is also identified as a significant multivariate future predictor in recent longitudinal research [14, 16], which is very consistent with self-report where “poverty” and “needing

⁵ Most of the following findings have also been recently reported in Williams et al. [16].

⁶ Gambling losses and gambling frequency tend to be stronger predictors relative to time spent gambling and total number of gambling activities engaged in. Having a large win in the past year is also predictive in some studies (e.g [6, 14]), as it is usually indicative of high levels of gambling expenditure (as well as likely being a predictor independent of its association with expenditure).

⁷ Thus, continuous forms of gambling could also be considered evidence of ‘intensive gambling’.

⁸ This is analogous to drug addiction, where the speed of the effect (e.g., injection or inhalation versus ingestion) and drug half-life are strong determinants of the dependency-forming potential of different substances [65].

money” are typically the most commonly self-identified reasons for gambling-related problems [6, 14, 16].

- *Male gender* and *younger age* are the most consistent individual correlates of gambling problems worldwide [1, 3], and male gender [5, 14] and younger age [16] have been implicated as predictors in some longitudinal studies. Their inconsistent *multivariate* relationship to future problem gambling is because they are strongly associated with stronger etiological factors (i.e., greater gambling intensity, impulsivity, EGM participation, gambling fallacies, peer group involvement, lower income) [1, 2].

An important caveat to the above etiological findings is that longitudinal studies have focused on the observable attributes and behaviour of the gambler, whereas a complete understanding of the etiology of problem gambling needs to recognize that between 40 and 60% of the propensity for developing these attributes/behaviours is due to genetic inheritance [91–94]. In a similar way, these studies provide insufficient focus on the contributing etiological role of how gambling is legally provided (i.e., types and amount of gambling available; harm minimization efforts) as this provision is the same for all gamblers within the jurisdiction/study (it requires cross-jurisdictional and/or before/after studies to identify their contribution). Gambling provision does play an important etiological role as evidenced by the fact that involvement in EGMs, casino table games, online gambling, and a greater number of gambling formats are all strong risk factors. A general review of the effectiveness of different harm minimization policies in preventing problem gambling is contained in Williams, West & Simpson [95].

Conclusions

Longitudinal research is necessary to understand the trajectory, chronicity, and etiology of problem gambling. Longitudinal studies of gambling and/or problem gambling were largely absent until 20 years ago. Since that time there have been over 50 such studies, with roughly 20 having a specific focus on problem gambling. The earliest studies were limited in scope, but later more comprehensive studies have rectified the weaknesses of this earlier research and have now collectively provided fairly definitive answers concerning trajectory, chronicity, and etiology.

First, problem gambling is not usually progressive. Rather, most people with early problems reduce their subsequent involvement and sustain this reduction. Second, problem gambling is also not usually chronic and unrelenting. A minority do have a sustained chronic course, particularly more severe cases. It is also enduring in that relapse and risk of relapse remains high in anyone who has ever had the condition. However, the condition most typically remits, with

some people having sustained remission and others having repeated relapses and remissions. Third, in terms of etiology, up to 20 variables additively contribute to the initial onset of problem gambling with different combinations of variables being present in different people. That said, intensive gambling involvement, gambling fallacies, addiction-propensity, a social group that regularly gambles, and impulsivity tend to be particularly important etiological factors. In general, problem gambling has a biopsychosocial etiology with a trajectory and chronicity very similar to substance use disorder.

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Recent papers in the past three years that are important (*) or very important (**):

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Declarations

Competing Interests The authors declare no competing interests.

Human and Animal Rights and Informed Consent All studies with human subjects are published and followed ethical standards.

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